

Pseudo-Factor Analysis with Item Intercepts

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Abstract

We extend the pseudo-factor analysis method by adding a mean structure to obtain pseudo-intercepts. We do so by substituting the item response mean vector based on human responses with an item mean vector obtained by projecting item embeddings via dot products onto semantic axes capturing the difference between low and high scale anchors. Across six scales measuring the trait domains of the DSM-5 alternative model for personality disorders in occupational settings, the mean correlation between empirical and pseudo-intercepts was $r = .54$, while the mean correlation between empirical and pseudo-intercepts obtained via deliberate projection onto randomly incorrect semantic axes was $r = -.35$. The inclusion of embedding-based means in pseudo-factor analysis can potentially allow mean and covariance matrix structure modeling without empirical data, opening the possibility of examining relative item locations, multiple-group pseudo-modeling, and measurement invariance analyses. Topics such as systematic methods to identify semantic anchors and risks including ‘anchor hacking’ are discussed.

Keywords: Pseudo-factor analysis, Intercepts

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Introduction

The use of text embeddings in psychological measurement has grown substantially in recent years following the emergence of transformer models in computer science (11) and demonstrations of relationships between item embedding associations and empirical item response correlations (4; 5; 13). Work has since shown that the structure of psychometric scales can be evaluated prior to empirical data collection (3; 1; 7; 9; 10; 12).

Embedding-Based Discriminations

Pseudo-factor analysis replaces the empirical item association matrix with a matrix of pairwise cosine similarities computed from item embeddings. Each item is embedded

using a sentence transformer model that produces a high-dimensional vector representation. Cosine similarity between item embedding pairs yields a symmetric similarity matrix with unit diagonal, structurally analogous to a correlation matrix rather than a covariance matrix.

This matrix is then factorized using exploratory factor analysis, which can be implemented as ESEM-within-CFA with fully specified oblique target rotation via the targetQ algorithm. Standardized pseudo-loadings that emerge have been shown to relate to standardized loadings estimated from observed item responses. However, pseudo-factor structures currently lack an explicit mean structure. We propose a proxy that can be substituted for the item mean vector, extending pseudo-factor analysis to include intercepts.

Role of Item Intercepts in Factor Analysis

Item intercepts are the expected item scores when the factor score is zero. They indicate where on the latent continuum the average respondent's response to an item sits. Higher intercepts indicate items that are more readily endorsed even by individuals low on the trait. Lower intercepts indicate items endorsed primarily by individuals high on the underlying trait.

Intercepts are required for full mean and covariance structure (MACS) analysis, including the comparison of latent means across groups and the detection of uniform measurement non-invariance (8). The absence of a mean structure in pseudo-factor analysis represents a meaningful gap in the method that we attempt to close with an embedding-based mean vector.

Embedding-Based Mean Vectors

Covariance structure analysis with a mean structure requires an explicit mean vector containing item means as input. This is absent in pseudo-factor analysis because there are no empirical responses to average. The rationale for the approach proposed here rests on the idea that a semantic axis or direction vector exists that represents the difference between low and high poles of psychometric scales. The axis is defined by the difference between embeddings of high and low anchor descriptions.

The pseudo-mean vector is obtained by dot product projection of item embeddings onto this semantic axis. The projection obtained per item is a scalar that is directly substituted for the item mean vector in factor model with the mean-structure. Items whose content aligns more closely with the high-pole anchor embedding receive larger projections and items closer to the low-pole anchor embedding receive smaller projections. To the extent that semantic proximity to the high pole reflects alignment with high-trait content, the projection may approximate relative item location along the latent continuum.

Because cosine similarity matrices do not contain empirical variance information, the present method uses the embedding similarity matrix in the role ordinarily occupied by the sample covariance matrix in SEM estimation. Although supplied to lavaan

through the `sample.cov` argument, the matrix is more appropriately interpreted as a standardized association matrix analogous to a correlation matrix. Estimation therefore proceeds without empirical item variances, using embedding-derived estimates of observed item means while treating the similarity matrix as a covariance substitute.

Method

Measures

We used the G-50, a brief DSM-5-based occupational measure of maladaptive personality traits (2), in this demonstration. The G-50 assesses six domain-level traits, each rated on a five-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Items were developed to capture mid-range maladaptive trait expression consistent with the DSM-5 alternative model but adapted for work settings.

Sample

We used the G-50 development sample, which comprised approximately 700 English-speaking working adults recruited online from across the United Kingdom and Ireland, with a mean age in the mid-thirties and representation across diverse industry sectors including finance, manufacturing, transport, and government.

Analysis

We embedded the item text using the all-mpnet-base-v2 sentence transformer, producing 768-dimensional vectors. Anchors were represented as adjectives capturing the high end and low end of each DSM-5 trait domain. We projected each item’s embedding onto direction vectors defined by the difference between the high and low anchor embeddings for its domain. Initial anchor definitions were chosen based on scale summaries and iteration via manual search. Three to four anchoring attempts per scale was sufficient to align the pseudo and empirical intercepts. The final anchors used appear in Table 1.

Table 1: Anchors used to define semantic difference axes

Domain	Low Anchor	High Anchor
Antagonism	Agreeable	Antagonistic
Compulsivity	Adaptable	Rigid
Detachment	Affectionate	Antisocial
Disinhibition	Organized	Impulsive
Emotionality	Secure	Troubled
Schizotypy	Normal	Weird

The item cosine similarity matrix was substituted for the covariance matrix. Embeddings and projections were computed in Python using the sentence-transformers

library. The cosine similarity matrix and pseudo-mean vector were then passed to R, where the mean-augmented pseudo-factor analysis was fit using the lavaan and esemComp packages.

Model Fit

Model fit for the empirical factor model was evaluated with χ^2 , CFI, TLI, RMSEA, and RMSR. Because pseudo-factor analysis has no empirical sample, fit indices that require a sample size—including χ^2 and any index derived from it such as CFI, TLI, and RMSEA—are not appropriate. Instead, we report residual-based fit, specifically the root mean square of the residuals (RMSR) between the model-implied and observed cosine similarity matrices. RMSR does not require a sample size and reflects how well the factor model reproduces the item similarity structure.

Criterion Validity

Criterion data for the pseudo-loadings and pseudo-intercepts were the empirical loadings and intercepts from an exploratory structural equation model of empirical response data to the same item set using maximum likelihood estimation with fully specified oblique target rotation via the targetQ algorithm, implemented using the esemComp package in R. The model was fit as an ESEM-within-CFA with mean structure, yielding standardized loadings and intercepts for each item from the empirical model.

Standardized loadings and intercepts were used because they place item location estimates on a common scale independent of the original response metric, making them comparable with embedding-based projections, which have no clear response scale. We compared the pseudo-means within each domain using Pearson correlations and scatter plots. As a baseline comparison, we deliberately projected item embeddings onto the wrong semantic direction axes by scrambling the low-high anchor pairs and re-running the analysis. We then compared the correlations between semantic and empirical intercepts to correlations when items were projected onto the correct semantic difference axes.

Results

Model fit for the empirical factor model was as follows: $\chi^2 = 2107.47$, $df = 940$; CFI = .90, TLI = .87, RMSEA = .042, and RMSR = .03. Model fit for the pseudo-factor analysis was assessed using RMSR given the inaccessibility of fit indices based on sample size. The RMSR for the six-factor ESEM solution was 0.048, falling below the conventional threshold of 0.05 that is often used for empirical data. This indicates that the factor model reproduced the observed similarity matrix with acceptable precision.

Loading Structure Evaluations

The empirical and semantic loading structures were evaluated using the dominant average absolute loading proposed in the original pseudo-factor analysis article to assign

factors, because factor assignment based only on simple counts of the number of items from a given scale with their highest loading on a factor is a blunt method that does not account for factor saturation. The DAAL analysis indicated that each factor was cleanly recovered. The factor structures are presented in Tables 2 and 3, with DAAL results in the final rows.

The final rows of Table 2 indicate that factor 1 is Negative Emotionality, factor 2 is Detachment, factor 3 is Schizotypy, factor 4 is Disinhibition, factor 5 is Compulsivity, and factor 6 is Antagonism. The final rows of Table 3 indicate that factor 1 is Schizotypy, factor 2 is Negative Emotionality, factor 3 is Compulsivity, factor 4 is Disinhibition, factor 5 is Detachment, and factor 6 is Antagonism. The factors, while recovered, emerged in different orders across empirical and pseudo-factor analysis, making this mapping step important for comparing factor loadings across methods.

Loading Criterion Validity

The correlations between empirical and semantic loadings ranged between .36 and .58, suggesting that semantic loadings are moderately informative about empirical loadings. Figure 1 shows that all items load on all factors consistent with an exploratory factor analysis, and the highest-loading items in both empirical and pseudo-factor analysis are the items intended to measure the target factor.

Intercept Criterion Validity

The pseudo-intercept structure was evaluated by checking whether the within-scale empirical intercepts were correlated with the within-scale semantic intercepts. Across the six domains, the correlations were all positive and ranged from moderate at .39 for Schizotypy to large at .70 for Disinhibition (Figure 2). The comparative baseline, where item embeddings were projected onto the wrong semantic axes, revealed intercept correlations between empirical and pseudo-standardized intercepts ranging from $-.71$ to $-.04$.

Discussion

This work proposed an extension to pseudo-factor analysis to obtain pseudo-intercepts by dot product projections of item embeddings onto a semantic axis representing the difference between embeddings of low and high anchor descriptions. Results indicate moderate to large positive correlations between semantic and empirical standardized intercepts, moderate correlations between empirical and pseudo-factor loadings, and adequate model fit using both approaches.

More research on pseudo-intercepts is needed before use in scale development and certainly in applied testing because claims regarding analyses it enables, such as semantic measurement invariance, have real consequences. This study was a proof-of-concept on a single measure with small within-domain item counts making systematic investigation of the method necessary. Because anchors were hand-tuned on intercepts,

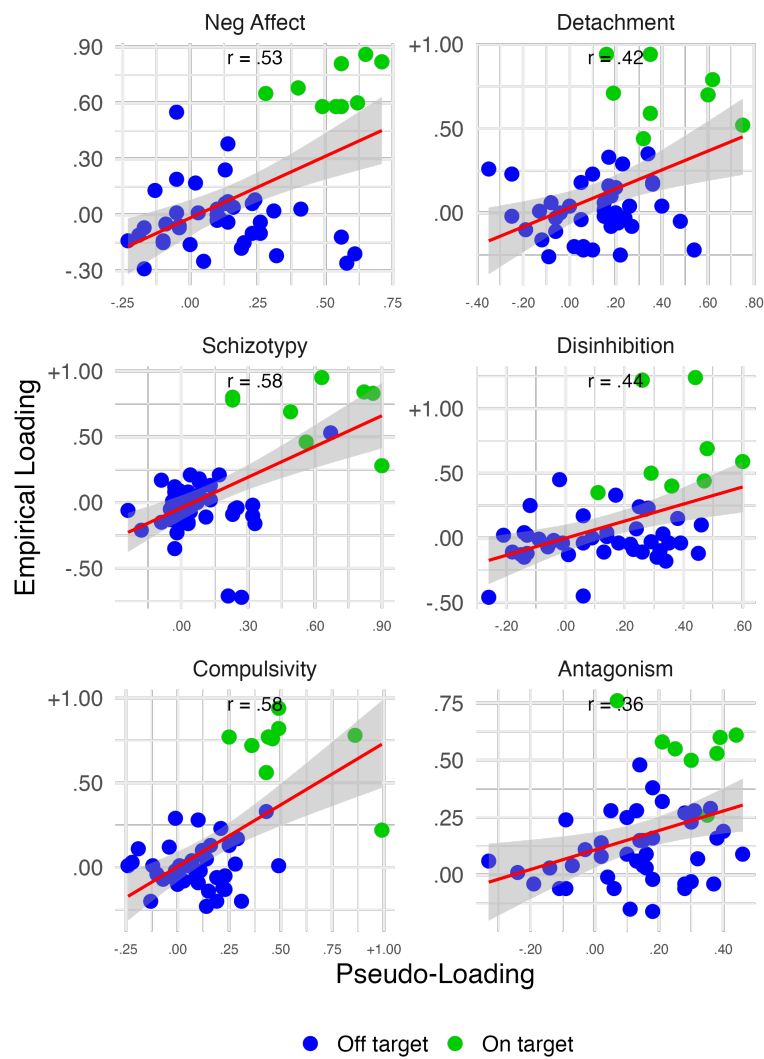


Figure 1: Pseudo-loadings vs. empirical loadings. Items loading on their target factor are highlighted.

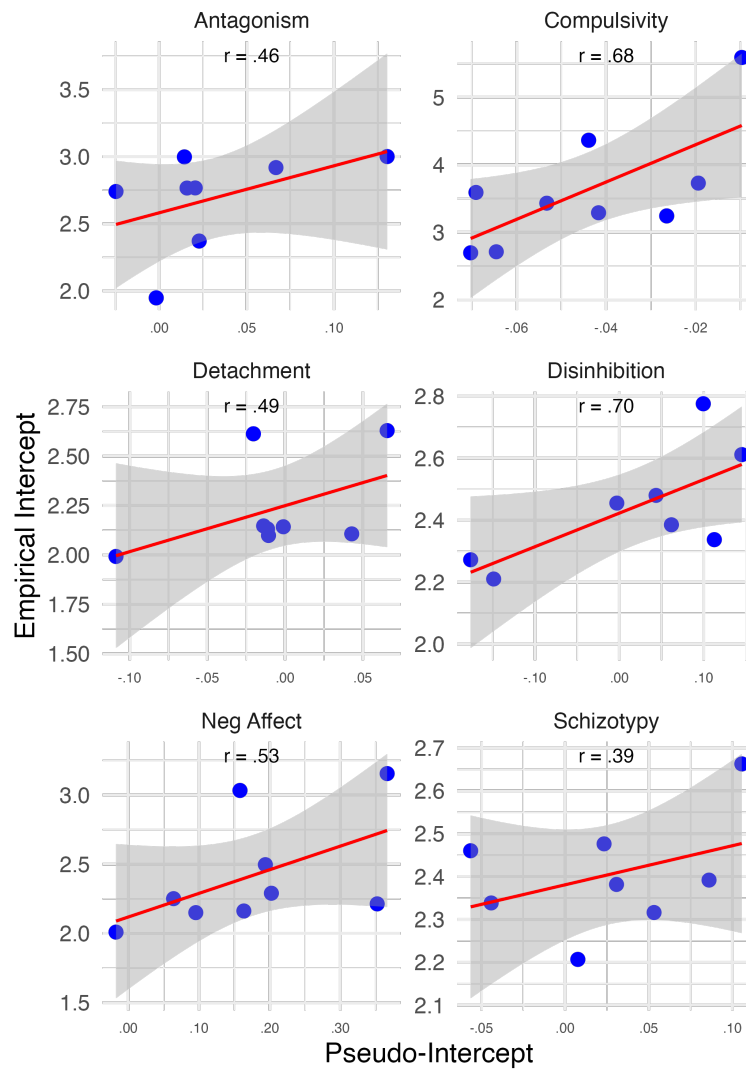


Figure 2: Pseudo-intercepts vs. empirical intercepts across the six DSM-5 trait domains.

correlations between intercept formats may be inflated, although the hand-tuning was limited to 3 or 4 attempts on each scale.

First, identifying appropriate low and high anchor poles for the difference vectors is a non-trivial hurdle that is susceptible to overfit due to *anchor hacking*. However, appropriate poles were identified for the proof of concept with relatively little experimentation, and iterative identification of factor markers is commonplace in psychometrics. Computational searches of appropriate adjective lists could further optimize the findings.

Second, the anchor directions identified here may not generalize across different embedding models, as the semantic geometry varies across encoders. A promising direction for future work would be to identify anchors that perform well for an existing dataset and then apply those anchors to new items to examine the transferability of anchors across models in pseudo-factor analysis with mean structures.

Future work should evaluate the robustness and generalizability of pseudo-intercepts through systematic ablation studies and cross-model validation. In particular, future analyses should quantify the sensitivity of pseudo-intercepts to anchor selection, embedding model choice, projection method, item wording perturbations, and comparisons against simpler baselines such as cosine similarity to the high anchor only.

Finally, an advantage of the approach described here is that it produces a unified geometric framework for the two key measurement parameters—loadings and intercepts—based on embedding geometry. However, how this approach compares to other methods of obtaining the mean vector, such as artificial crowd responses or predictive machine learning models of item means, requires further attention.

References

- [1] Guenole, N.; D’Urso, E. D.; Samo, A.; Sun, T.; and Haslbeck, J. 2025. Enhancing scale development: Pseudo-factor analysis of language embedding similarity matrices. *PsyArXiv*.
- [2] Guenole, N. 2015. The hierarchical structure of work-related maladaptive personality traits. *European Journal of Psychological Assessment*, 31(2):83–90.
- [3] Guenole, N.; Samo, A.; and Sun, T. 2024. Pseudo-factor analysis of embedding similarity matrices. Paper presented at the 2024 International Meeting of the Psychometric Society, Prague, Czech Republic.
- [4] Hernandez, I.; and Nie, W. 2023. The AI-IP: Minimizing the guesswork of personality scale item development through artificial intelligence. *Personnel Psychology*, 76(4):1011–1035.
- [5] Hommel, B. E.; and Arslan, R. C. 2025. Language models accurately infer correlations between psychological items and scales from text alone. *Advances in Methods and Practices in Psychological Science*, 8(4).
- [6] Lorenzo-Seva, U.; Timmerman, M. E.; and Kiers, H. A. 2011. The Hull method for selecting the number of common factors. *Multivariate Behavioral Research*, 46(2):340–364.
- [7] Milano, N.; Luongo, M.; Ponticorvo, M.; and Marocco, D. 2025. Semantic analysis of test items through large language model embeddings predicts a-priori factorial structure of personality tests. *Current Research in Behavioral Sciences*, 8:100168.
- [8] Sörbom, D. (1974). A general method for studying differences in factor means and factor structure between groups. *British Journal of Mathematical and Statistical Psychology*, 27(2), 229–239.
- [9] Suárez-Álvarez, J.; He, Q.; Guenole, N.; and D’Urso, D. 2026. Using artificial intelligence in test construction: A practical guide. *Psicothema*, 38(1):1–12.
- [10] Varrasi, S.; Platania, G. A.; Castellano, S.; Ferraioli, F.; Massimino, S.; Vicario, C. M.; and D’Urso, E. D. 2026. Expanding psychometrics with pretrained language models: Evaluating pseudo-factor analysis in applied and multilingual contexts. *Methods in Psychology*, 100244.
- [11] Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- [12] Voss, N. M.; Wu, F. Y.; Javalagi, A. A.; and Kell, H. J. 2026. Integrating ensemble clustering and text embeddings for estimating the factor loadings of self-report scales. *Educational and Psychological Measurement*.

- [13] Wulff, D. U.; and Mata, R. 2025. Semantic embeddings reveal and address taxonomic incommensurability in psychological measurement. *Nature Human Behaviour*, 9(5):944–954.

Appendix

Table 2: Empirical Standardized Loading Structure

Item	NE	DE	SC	DI	CO	AN
ANHO54	.01	.06	.08	.23	.13	.58
ANAG19	.04	.18	.08	.17	.13	.53
ANMN51	.03	.33	.00	.25	.02	.26
ANCA14	.08	.02	.09	.33	.29	.76
ANHS17	.07	.08	.06	.24	.17	.61
ANNC16	.04	.06	.09	.45	.28	.50
ANOP92	.06	.18	.06	.10	.33	.60
ANDC57	.04	.06	.05	.22	.23	.55
COPF27	.13	.00	.17	.10	.76	.29
COOR102	.15	.22	.16	.45	.77	.06
CORG29	.05	.02	.07	.15	.94	.23
COPS64	.07	.02	.23	.11	.82	.27
CORA67	.29	.21	.21	.18	.56	.28
COPF63	.03	.01	.16	.01	.72	.32
CORG101	.07	.04	.00	.02	.22	.24
COPS100	.06	.16	.21	.12	.77	.19
COCP26	.14	.20	.13	.46	.78	.04
DESW10	.55	.44	.13	.11	.05	.16
DESD11	.15	.79	.01	.05	.09	.07
DERA12	.12	.59	.17	.11	.01	.15
DESW82	.22	.94	.04	.13	.06	.01
DESW46	.10	.52	.09	.09	.02	.03
DEAH13	.21	.94	.06	.15	.09	.06
DERA84	.03	.71	.02	.04	.00	.09
DESD83	.26	.70	.08	.04	.01	.04
DIDT95	.02	.26	.72	1.24	.02	.09
DIRK60	.02	.17	.71	1.22	.01	.08
DIIR97	.25	.23	.15	.44	.13	.28
DIIM94	.11	.26	.53	.35	.06	.25
DIDT59	.17	.23	.10	.40	.12	.16
DIRK96	.19	.29	.35	.69	.20	.15
DIIR61	.24	.15	.01	.50	.14	.48
DIIM58	.01	.11	.12	.59	.01	.38
NEDP44	.60	.22	.18	.13	.10	.15
NELS78	.86	.10	.05	.07	.06	.09
NEPE41	.68	.25	.21	.03	.05	.04
NEAX74	.65	.04	.04	.04	.04	.04
NEGS7	.82	.04	.15	.00	.07	.06
NEEM73	.81	.16	.11	.12	.04	.06
NESP9	.58	.22	.02	.04	.12	.16

Item	NE	DE	SC	DI	CO	AN
NESE40	.58	.05	.01	.01	.11	.02
NESB75	.58	.35	.16	.02	.03	.28
SCUP104	.10	.03	.84	.03	.10	.03
SCCD71	.38	.10	.28	.15	.01	.01
SCUB69	.00	.04	.46	.07	.20	.06
SCEC34	.16	.03	.69	.02	.04	.03
SCDS108	.03	.00	.80	.13	.23	.11
SCUP32	.18	.06	.83	.04	.20	.06
SCEC106	.14	.08	.95	.04	.08	.04
SCDS72	.06	.20	.78	.04	.06	.14
AN	.05	.12	.06	.25	.20	.55
CO	.11	.10	.15	.18	.70	.21
DE	.21	.70	.08	.09	.04	.08
DI	.13	.21	.34	.68	.09	.23
NE	.68	.16	.10	.05	.07	.10
SC	.13	.07	.70	.07	.12	.06

Table 3: Pseudo-Factor Standardized Loading Structure

Item	SC	NE	CO	DI	DE	AN
ANHO54	.03	.03	.25	.28	.18	.21
ANAG19	.01	.14	.16	.06	.36	.38
ANMN51	.07	.10	.28	.12	.17	.35
ANCA14	.23	.24	.01	.17	.15	.07
ANHS17	.08	.04	.29	.25	.27	.44
ANNC16	.00	.26	.10	.02	.21	.30
ANOP92	.24	.04	.43	.46	.05	.39
ANDC57	.05	.16	.21	.27	.15	.25
COPF27	.01	.13	.46	.32	.20	.36
COOR102	.08	.10	.25	.06	.06	.28
CORG29	.04	.09	.49	.31	.15	.30
COPS64	.02	.14	.49	.26	.25	.28
CORA67	.17	.17	.43	.34	.05	.13
COPF63	.03	.10	.36	.14	.13	.21
CORG101	.01	.17	.99	.21	.05	.09
COPS100	.18	.13	.44	.45	.12	.40
COCP26	.13	.23	.86	.26	.02	.07
DESW10	.05	.05	.23	.13	.32	.18
DESD11	.04	.20	.10	.22	.62	.32
DERA12	.09	.56	.05	.18	.35	.15
DESW82	.07	.32	.09	.15	.35	.04
DESW46	.01	.26	.01	.23	.75	.16
DEAH13	.01	.61	.10	.14	.16	.06
DERA84	.32	.10	.02	.14	.19	.46

Item	SC	NE	CO	DI	DE	AN
DESD83	.03	.58	.24	.14	.60	.15
DIDT95	.27	.11	.11	.44	.35	.10
DIRK60	.21	.31	.12	.26	.36	.02
DIIR97	.09	.05	.23	.47	.10	.31
DIIM94	.67	.19	.19	.11	.09	.10
DIDT59	.32	.02	.22	.36	.25	.38
DIRK96	.03	.05	.31	.48	.23	.14
DIIR61	.02	.13	.15	.29	.20	.14
DIIM58	.03	.05	.49	.60	.06	.18
NEDP44	.08	.62	.12	.15	.10	.11
NELS78	.24	.65	.01	.06	.18	.16
NEPE41	.04	.40	.14	.29	.22	.28
NEAX74	.25	.28	.10	.39	.26	.19
NEGS7	.03	.71	.07	.09	.40	.11
NEEM73	.11	.56	.09	.13	.17	.13
NESP9	.13	.56	.04	.01	.54	.18
NESE40	.02	.49	.19	.09	.48	.18
NESB75	.33	.54	.22	.04	.34	.05
SCUP104	.82	.23	.00	.33	.06	.14
SCCD71	.90	.14	.01	.38	.19	.24
SCUB69	.56	.10	.19	.24	.00	.09
SCEC34	.49	.00	.07	.13	.24	.30
SCDS108	.23	.41	.14	.01	.04	.03
SCUP32	.86	.19	.13	.35	.08	.33
SCEC106	.63	.10	.03	.06	.18	.37
SCDS72	.23	.23	.02	.18	.06	.02
AN	.09	.13	.22	.20	.19	.30
CO	.07	.14	.53	.26	.11	.23
DE	.08	.33	.10	.17	.42	.19
DI	.20	.11	.23	.38	.20	.17
NE	.13	.53	.11	.14	.30	.15
SC	.59	.18	.07	.21	.11	.19